Biology — Detailed Marking Report

Federal Board Annual Examination 2026, Version 3 | Chapters 1-7 (Ch 7: Sec 7.1-7.6)
Candidate: Seemab | Attempt Date: 6 July 2026 | Marked: 6 July 2026 | Exam: exam-09-ch1-7

61.25/65
Total Score

94.2%
Percentage

A+
Grade

Section A (MCQ): **12/12** PERFECT — 13th consecutive
Section B (Short): **32/33**
Section C (Long): **17.25/20**

Best full-syllabus Ch1-7 Biology paper to date. 94.2% beats exam-07-ch1-7 (88.5%) by 5.7 points and exam-08-ch1-7 (83.8%, the immediately prior attempt) by 10.4 points. Two questions initially missing from the first scan — 2(ii) and 2(iii) — were recovered from a complete 12-page re-scan and both scored full marks; see recovery notes below.

SECTION A — Multiple Choice Questions (12/12)

Q	Seemab's Answer	Correct Answer	Result
1	D	**D**	✓
2	B	**B**	✓
3	A	**A**	✓
4	D	**D**	✓
5	D	**D**	✓
6	B	**B**	✓
7	A	**A**	✓
8	C	**C**	✓
9	C	**C**	✓
10	B	**B**	✓
11	C	**C**	✓
12	A	**A**	✓

All 12 MCQs correct — Seemab's strongest MCQ page in Biology to date. This is the 13th consecutive perfect MCQ score across all subjects (attempts 14–26), maintaining a 99.4% all-time MCQ record (310/312).

SECTION B — Short Answer Questions (32/33)

Q.2(i) OR Three Muslim scientists and the three major fields of biology — **3/3**

Seemab wrote: Jabir bin Hayyan, Abdul Malik Asmai, and Bu Ali Sina named as the three Muslim scientists who contributed to the present-day knowledge of plants and animals. Three major fields correctly stated: Botany (study of plants), Zoology (study of animals), Microbiology (study of micro-organisms).
Complete and precise — both halves of the OR variant covered in full.

Q.2(ii) OR Autotrophic, heterotrophic, and saprotrophic nutrition — **3/3**

Seemab wrote: all three modes of nutrition correctly defined and matched — autotrophic (photosynthetic, e.g. green plants), heterotrophic (ingestive, e.g. animals), saprotrophic (absorptive, e.g. fungi/bacteria).
Full marks — the ingestive-vs-absorptive distinction, a detail many students blur, was captured correctly.

> Recovered from the 6 July re-scan — absent from the first (incomplete) 11-page scan.

Q.2(iii) OR Taxonomic hierarchy of the Pea plant — **3/3**

Seemab wrote: all seven ranks correct — Kingdom Plantae, Division Magnoliophyta, Class Magnoliopsida, Order Fabales, Family Fabaceae, Genus Pisum, Species Pisum sativum.
Full marks — complete hierarchy, correctly ordered.

> Recovered from the 6 July re-scan — absent from the first (incomplete) 11-page scan.

Q.2(iv) main Three layers of the plant cell wall + plasmodesmata — **3/3**

Seemab wrote: middle lamella (pectin, holds neighbouring walls together), primary wall (criss-cross cellulose fibres),

secondary wall (lignin-thickened, e.g. xylem vessels) all correctly described; plasmodesmata correctly defined as pores forming cytoplasmic connections between neighbouring cells.
Complete, textbook-precise answer.

Q.2(v) OR Ribosome structure and prokaryotic vs eukaryotic differences **3/3**

Seemab wrote: ribosomes made of two subunits (one large, one small) joining to function; non-membranous granular structures found in both prokaryotic and eukaryotic cells; ~55% of cell dry weight is protein; prokaryotic ribosomes smaller than eukaryotic.
All three required parts of this OR variant covered correctly.

Q.2(vi) OR Chromatid, centromere, and sister chromatid formation in S-phase **3/3**

Seemab wrote: chromatid defined as one of two identical copies of a chromosome formed after DNA replication; centromere defined as the point joining sister chromatids; S-phase correctly explained as the replication step that produces two sister chromatids per chromosome.
Full marks, clean and complete.

Q.2(vii) main Lymphatic and immune systems — organs + homeostatic role **3/3**

Seemab wrote: Lymphatic system organs (lymph, lymph nodes, lymph vessels) with role of defending against infection and maintaining tissue fluid balance; Immune system organs (leukocytes, tonsils, adenoids, thymus, spleen) with role of removing pathogens and fighting infection.
Both systems fully covered with correct organs and roles.

Q.2(viii) OR Cardiovascular and skeletal systems — organs + homeostatic role **3/3**

Seemab wrote: Cardiovascular system (heart, blood, blood vessels) stabilizing gases, wastes, temperature and pH; Skeletal system (bones, cartilage, joints, tendons, ligaments) regulating calcium and mineral levels in blood.
Complete and correct.

Q.2(ix) main Nervous and respiratory systems — organs + homeostatic role **3/3**

Seemab wrote: Nervous system organs (brain, spinal cord, nerves, sensory organs) controlling rapid short-term change; Respiratory system organs (nose, pharynx, larynx, trachea, bronchi, lungs, diaphragm) balancing oxygen and carbon dioxide.
Full marks — both systems named and functionally explained.

Q.2(x) OR Ten trace (minor) bioelements **3/3**

Seemab wrote: all ten trace bioelements listed — Potassium, Sulphur, Chlorine, Sodium, Magnesium, Iron, Copper, Manganese, Zinc, Iodine.
Complete list, full marks.

Q.2(xi) main Metabolic pathway and the role of enzymes **2/3**

Seemab wrote: "Many enzymes work together in a specific sequence to make metabolic pathways. A metabolic pathway is a series of connected chemical reactions." Both of the first two required points are present and correct.

> Deduction −1: The third point is missing entirely — that multiple intermediate molecules are produced before the initial substrate is converted into the final product. This is the "assembly line" detail that distinguishes a pathway (many steps) from a single reaction, and it carries its own mark in the scheme.

SECTION C — Long Answer Questions (17.25/20)

Q.3(i) main Five sub-fields of biology with example/application **4.5/5**

Seemab wrote: Histology (microscopic study of tissues), Genetics (study of genes and heredity in organisms), Embryology (study of the development of an organism from a fertilized egg), Paleontology, and Immunology (the study of immunity, the ability of the body to protect itself from substances like microbes) — five sub-fields named, as required.

> Deduction −0.5: Paleontology is defined as "the study of the history of life on Earth based on fossil fuels" — it should read "fossils", not "fossil fuels". A one-word slip that changes the meaning of the definition. The examples/applications given alongside each sub-field were thin (mostly definitions only, without a distinct worked example) but were accepted per the answer key's flexible phrasing, so no further deduction was applied for that.

Q.3(ii) OR Plant vs animal cell advantages + stem cells **3/5**

Seemab wrote: a complete five-point comparison table — cell wall cementing adjoining cells to form a supportive structure vs support from collective tissue/organ/skeletal arrangement; xylem/phloem transport channels vs transport through different structures and mechanisms; rigid cell wall withstanding osmotic stress and storing water vs the animal cell's inability to do so; turgidity keeping the plant upright vs no turgidity in animal cells; immobility due to cell wall rigidity vs animal cell flexibility to move and find shelter, food, or better reproductive opportunities. All five points correct — full marks on this half (3 of 5).

Deduction −2: The second half of this question — describing stem cells with reference to haematopoietic stem cells — was not attempted at all. No mention of the zygote as the most basic stem cell, and no mention of haematopoietic stem cells producing red blood cells, white blood cells, platelets, and other cell types. The answer moves directly from the comparison table into Q.3(iii) with nothing written for this half, which is worth 2 of the 5 marks.

Q.3(iii) main Eight differences between mitosis and meiosis **4.75/5**

Seemab wrote: a full eight-row comparison table — occurs in animal cell / germ line cells; cell divides once / twice; produces 2 / 4 daughter cells; diploid / haploid daughter cells; daughter cells become part of the somatic body / form gametes; chromosomes remain independent / homologous chromosomes pair (synapsis); no chiasmata / chiasmata formed; no crossing over / crossing over during meiosis I. All eight required rows present.

Deduction −0.25: Row 1's mitosis side reads "Occurs in animal cell" — the precise textbook contrast is "somatic cells" vs "germ line cells" (mitosis is not limited to animal cells). The other seven rows are fully correct. A small terminology slip on an otherwise excellent table.

Q.3(iv) main Lipid structure, saturated vs unsaturated fatty acids, two functions **5/5**

Seemab wrote: complete coverage of fatty acids + glycerol forming acylglycerides, saturated fatty acids (all C–C single bonds, solid at room temperature, e.g. cheese/butter/ghee) vs unsaturated fatty acids (double/triple C–C bonds, liquid at room temperature, e.g. soybean/mustard/olive oil), Function 1 (phospholipids and cholesterol as components of the plasma membrane), Function 2 (lipids as an energy store, 9.1 kcal/g).
Full marks — complete and precise across every required part.

DEDUCTION SUMMARY

Question	What was missing / wrong	Marks lost
2(xi) main	Third point of metabolic pathway definition missing (intermediate molecules produced before final product)	−1
3(i) main	Paleontology defined using "fossil fuels" instead of "fossils"	−0.5
3(ii) OR	Stem cells / haematopoietic stem cells half entirely omitted	−2
3(iii) main	Mitosis row 1: "animal cell" instead of "somatic cells"	−0.25
	Total deducted	**−3.75**
	Final score: 65 − 3.75 =	**61.25**

Strengths

- Perfect MCQs (12/12) — Seemab's strongest MCQ page in Biology to date; 13th consecutive perfect MCQ across all subjects.
- 10 of 11 Section B questions scored full marks, all with textbook-precise wording.
- Both questions recovered from the re-scan (2(ii), 2(iii)) scored full marks — no content gap, only a scanning gap.
- Lipid structure (Q.3(iv)) — 5/5, complete structural and functional coverage.
- Mitosis vs meiosis comparison (Q.3(iii)) — 7 of 8 rows flawless, only one terminology slip.

Areas to Improve

- Two-part long questions: answer BOTH halves. Q.3(ii) lost 2 of 5 marks purely from skipping the stem-cells half after finishing the comparison table.
- Include every key point in short answers even when the main idea is captured — Q.2(xi) needed a third point (intermediate molecules) that was never written.
- Small factual precision: "fossils" not "fossil fuels" (Paleontology, Q.3(i)); mitosis occurs in "somatic cells", not "animal cells" (Q.3(iii)).

Watch-Item: Stopping Before the Second Half of a Two-Part Question

This is the same "stopping before completing every required sub-part" pattern that has recurred across Physics, Chemistry, and Math over the last several exams (most recently broken by Math exam-11's second attempt on 5 July). It resurfaces here in Biology on Q.3(ii): the five-point plant-vs-animal comparison was completed cleanly, but the stem-cells half was never started. At 2 of 3.75 total marks lost, this single omission is the largest deduction on the paper. Drill fix: before writing an answer, re-read the full question and confirm how many distinct parts it is asking for (here: "compare cells" **and** "describe stem cells") before considering it finished.

Top Revision Targets

- **Metabolic pathways (Ch 7):** A pathway is a *series* of connected reactions — always add the "multiple intermediate molecules before the final product" point, not just the definition.
- **Stem cells (Ch 3):** Zygote = the most basic stem cell (source of ~220 cell types); haematopoietic stem cells produce red blood cells, white blood cells, platelets, and more — memorise this as its own mini-topic, not an afterthought to the plant-vs-

animal comparison.

- **Paleontology (Ch 1):** Study of the history of life on Earth based on **fossils** (not fossil fuels).
- **Mitosis vocabulary (Ch 4):** Mitosis occurs in **somatic cells**; meiosis occurs in germ line cells. Avoid the shorthand "animal cell".

COMPARISON WITH PRIOR BIOLOGY ATTEMPTS

Date	Exam	Score	Percentage	MCQ	Grade
3 Apr 2026	exam-01-ch1-6	52/65	80.0%	11/12	A
30 Apr 2026	exam-02-ch7 (1st)	60.5/65	93.1%	12/12	A
2 May 2026	exam-02-ch7 (re-sit)	65/65	100.0%	12/12	A+
8 Jun 2026	exam-03-ch4	62.5/65	96.2%	12/12	A+
23 Jun 2026	exam-07-ch1-7	57.5/65	88.5%	12/12	A
29 Jun 2026	exam-08-ch1-7	54.5/65	83.8%	12/12	A
6 Jul 2026	**exam-09-ch1-7 (this)**	**61.25/65**	**94.2%**	**12/12**	**A+**

Biology subject average (7 attempts): 413.25/455 = 90.8%. 94.2% is Biology's 3rd-highest score to date (behind only the 100% exam-02 re-sit and the 96.2% exam-03-ch4), and by far the best of the three comprehensive Ch1-7 papers — a +5.7 point improvement over exam-07 (88.5%) and a +10.4 point improvement over the immediately prior attempt, exam-08 (83.8%).